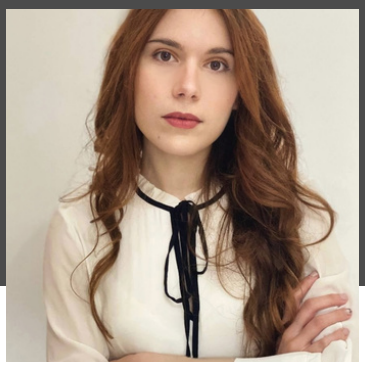


LIDIA SCUDIERO

DATA SCIENCE |NEUROSCIENCE | MACHINE LEARNING| DECISION SUPPORT|EXPLAINABLE AI

PROFILE

Big Data and AI graduated, with experience in data science and applied research. I build analytics pipelines from data cleaning and feature engineering to dashboards and report. I am particularly interested in using data, AI and behavioural insights to support evidence-based decisions in complex technical environments.

CONTACT INFORMATION

E-mail: scudierolidia@gmail.com
LinkedIn: lidia scudiero_linkedin
Github: lidia scudiero_github
Portfolio: lidia scudiero_portfolio

SELECTED IMPACT

- **Challenged an established assumption with real-world data analysing 7.5M IoT telemetry records and cross-referenced** them with technical documentation, **finding that 99% of recorded failures appeared as sudden events**, prompting a review of whether predictive modeling was appropriate.
- **Connected fragmented sources** linking attack datasets, Reddit discussions, scientific literature, and EU funding information using knowledge graphs, RAG, time-series, and geospatial analysis to **investigate relationships between technology, public perception, and research priorities.**
- **Recognised for academic excellence** selected by the **Italy-USA Foundation** among outstanding Italian graduates in Cognitive Neuroscience; preselected for the **ESA Graduate Trainee Programme** (Data Quality), 2026.

PROFESSIONAL EXPERIENCE

LLM EVALUATION AND QUALITY SPECIALIST| DataAnnotation Tech | Remote| May 2026 - Present

- **Evaluate** LLM outputs against defined quality and **reasoning criteria**, **identifying biases,** inconsistencies, and **instruction-following failures**.
- Apply **structured evaluation** and **prompt-engineering** principles to support systematic **quality control** and **evidence-based** assessment of AI outputs.

DATA SCIENCE INTERN | Almawave S.p.A.| Remote|Sep 2025 - Dec 2025

- Conducted a **feasibility study** for **predictive maintenance of IoT-enabled ticket gates**, analysing **7.5M telemetry records** to assess failure patterns, data quality, and model applicability.
- **Found** that approximately **99% of recorded failures appeared as sudden events**. Cross-referenced telemetry with **technical documentation** and investigated whether the discrepancy reflected the underlying system, available signals, or how failures were defined and recorded.
- **Built an end-to-end analytical pipeline** for data cleaning, KPI computation, anomaly/outlier detection, and exploratory analysis with **machine learning and deep learning models.**
- Translated findings and modelling limitations into **interactive dashboards and reports** for technical and non-technical stakeholders, and proposed alternative monitoring and data-governance strategies.

CIVIL SERVICE VOLUNTEER| Social Services Office, Giffoni Sei Casali| June 2023 - June 2024

- Provided **digital assistance and front/back office** support, collecting and managing **user data for social projects.**
- Supported vulnerable people and **contributed to basic impact analysis activities,** strengthening teamwork, communication and problem-solving skills.

WEB EDITOR AND PRIVATE TUTOR| Salerno, Mondadori S.p.A-formerly Banzai media o2.o.it| 2015-2024

- **Researched** and wrote online articles on **technology, science, and education, translating complex topics into clear and accessible** content for non-expert audiences.
- Developed strong r**esearch, synthesis, and communication skills** through **independent topic research** and long-term **writing experience**.
- Delivered private lessons to primary and secondary students, **strengthening communication and audience-adaptation skills**.

SELECTED PROJECTS (more on my portfolio and github)

Motor Imagery BCI - EEG Decoding

- Developed an **end-to-end EEG decoding pipeline** for **motor-imagery BCI**, covering preprocessing, spatial filtering, and machine/deep-learning classification.
- Evaluated **signal quality, interpretability, and subject-independent generalisation, investigating limitations relevant to robust BCI applications**.

Alzheimer's Clinical Support System

- Developed an **interpretable clinical decision-support system** for dementia-risk stratification using heterogeneous clinical and lifestyle data.
- Investigated class imbalance and demographic bias, compared tree-based models, and used SHAP to support **evidence-based interpretation of model predictions**.

Cybersecurity Analytics and Multi-Source Research

- Combined attack datasets, Reddit discussions, scientific literature, and **EU funding information to investigate relationships between cyberattacks, public perception, and research priorities.**
- Used knowledge graphs, RAG, entity extraction, time-series, and geospatial analysis to **connect fragmented information and identify patterns across heterogeneous sources.**

TECHNICAL SKILLS

Programming: Python, SQL
Data analysis and BI: Advanced Excel, Power BI,R, SPSS, Visual Studio, Streamlit, DuckDB
Data science Tools : Pandas, Numpy, Scipy, Matplotlib, Seaborn, Altair, Folium, Plotly, scikit-mobility, Shapely, OSMnx, networkx
ML and AI : Scikit-Learn, Keras, TensorFlow, sktime, NLTK, Spacy(NLP)
Neuroscience and Signal Processing: MNE-Python, PyLSL, Braindecode, pyRiemann
Web and Data Crawling: Selenium, Beautiful Soup
Generative AI: Hugging Face, Mistral, Cohere
Collaboration and other tools: Notion, Docker, Git, Canva, Inkscape, Psytoolkit, Social media

LANGUAGES

Italian - Native, English -Advanced

EDUCATION

Second-level Master in Big Data Analytics and Artificial Intelligence for Society | University of Pisa| 2024-2026

- Final Grade: Excellent
- Data mining, machine learning, data visualization, data journalism, web scraping, NLP, information retrieval, social network analysis.

MSc in Cognitive Psychology | University of Campania "Luigi Vanvitelli" | 2020-2023

- Curriculum : Cognitive Neuroscience
- Final Grade: 110 cum Laude

BSc in Psychological Sciences and Techniques | University of Campania "Luigi Vanvitelli" | 2017-2020

- Final Grade: 110 cum Laude

Merit-based Scholarship (2017–2021) for academic excellence and high academic performance during university studies.